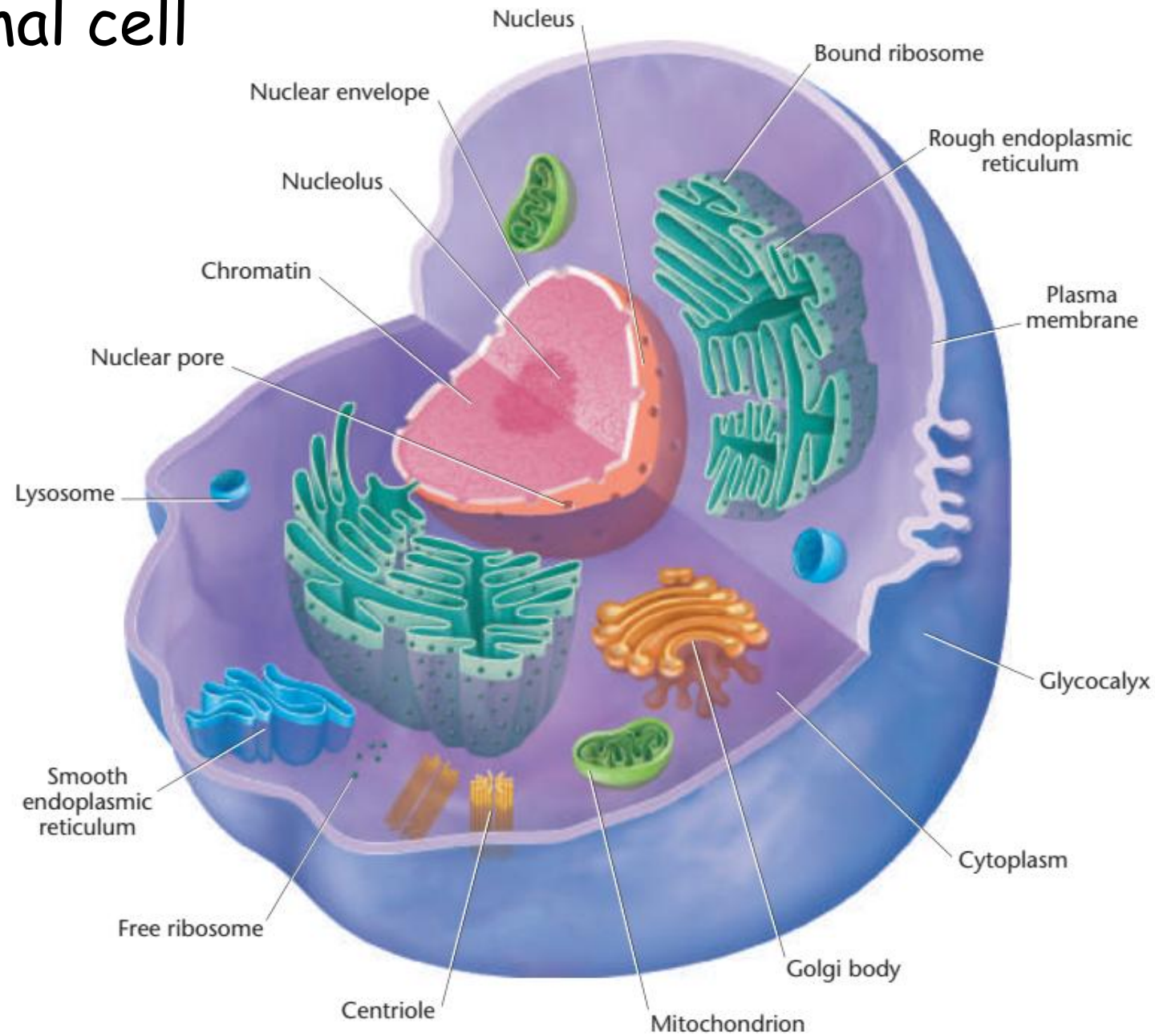
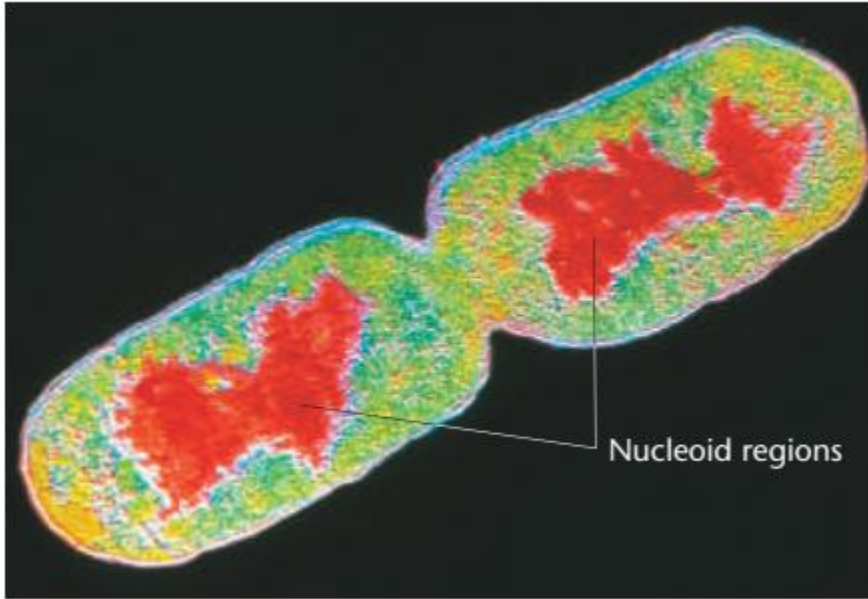


Cell Division and its control

Dr.Ayan Sadhukhan
BSBE, IIT Jodhpur

A general animal cell

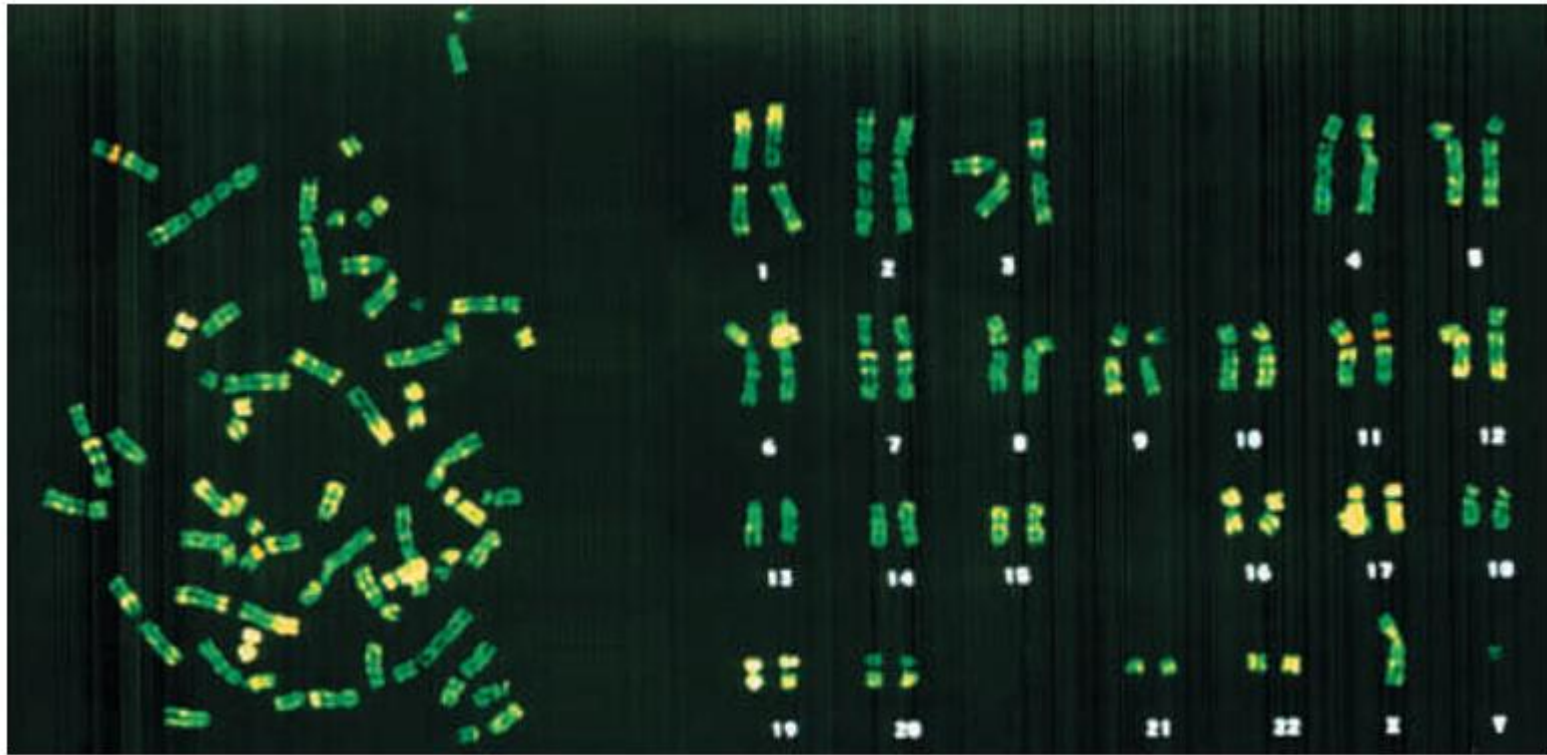




No nucleus, but naked DNA
as nucleoid (red)

Color-enhanced electron micrograph
of a bacterium undergoing cell division

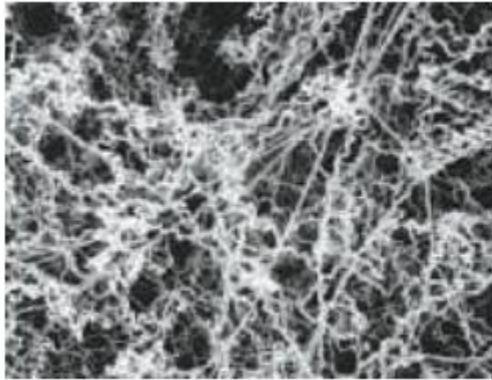
In diploid eukaryotes, chromosomes exist in pairs (homologous chromosomes)



A karyotype of a male human,
46 (23 pairs) of metaphase chromosomes

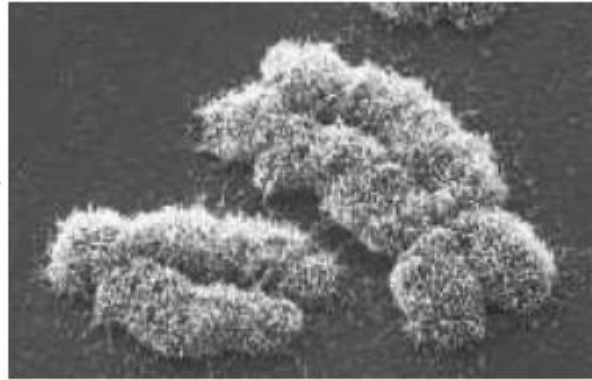
We got 23
from our
father, 23
from our
mother

Electron micrographs of eukaryotic DNA in different stages of the cell cycle



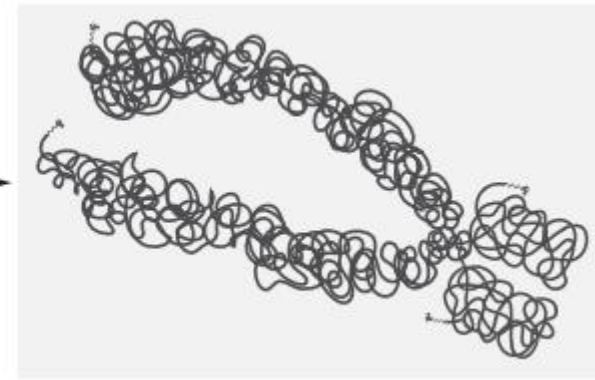
Interphase

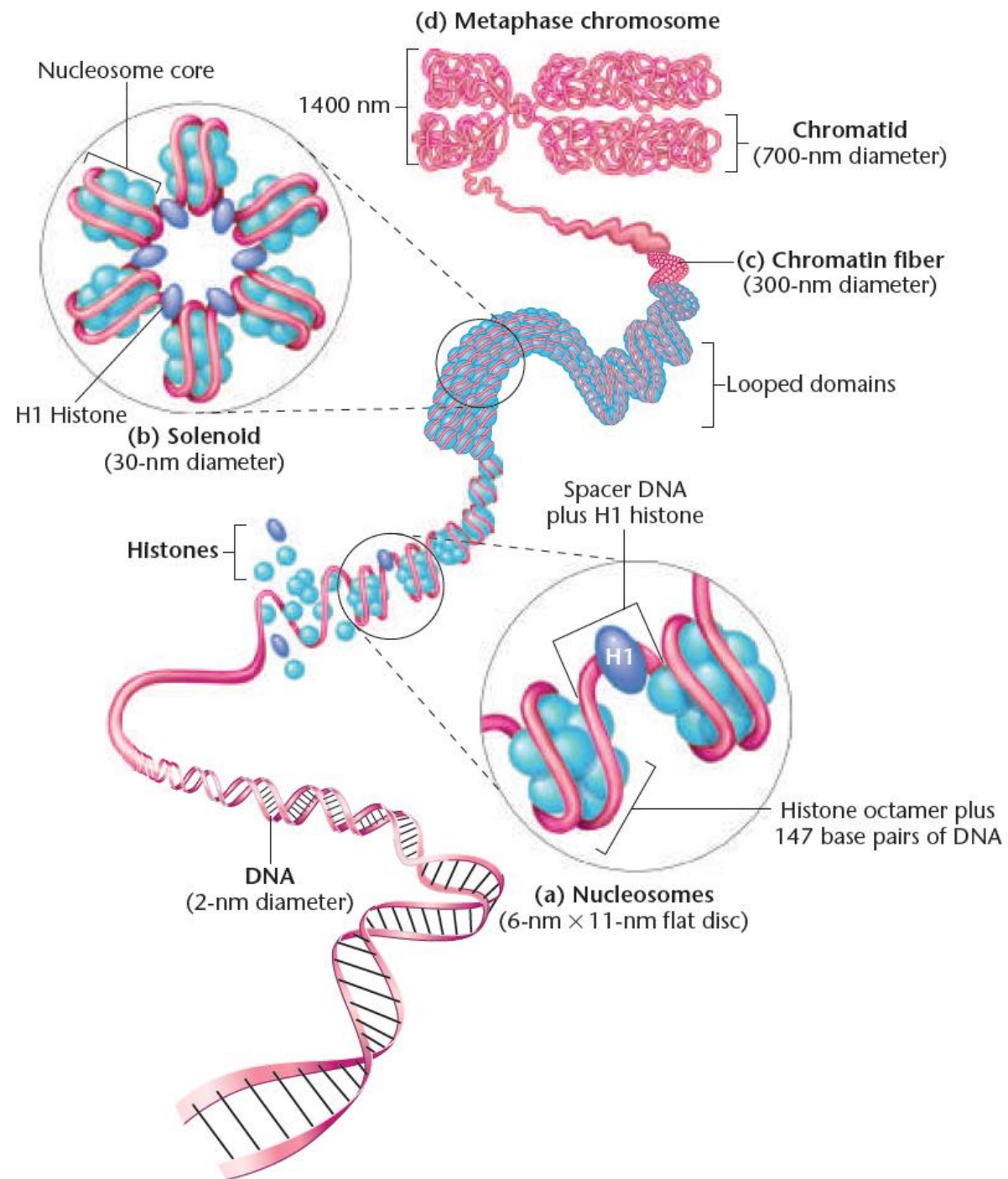
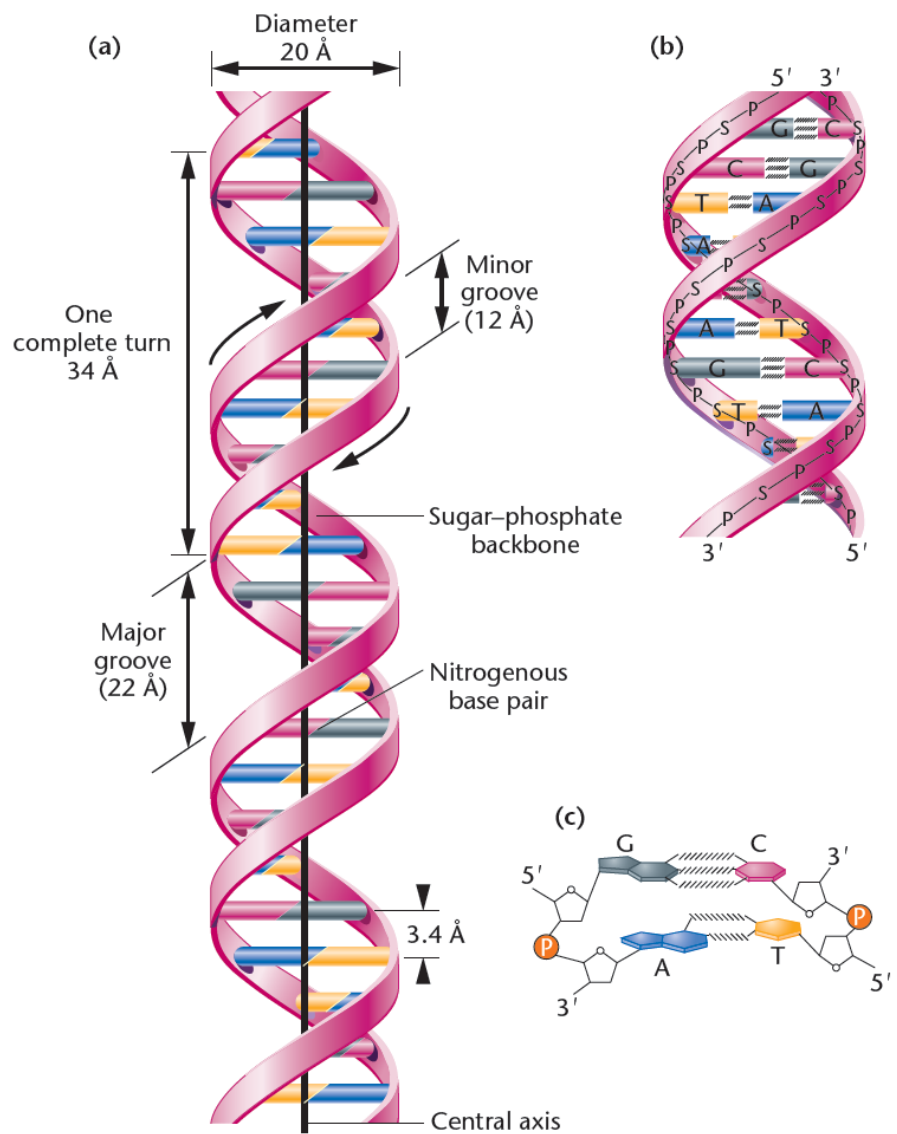
DNA is seen
as strands



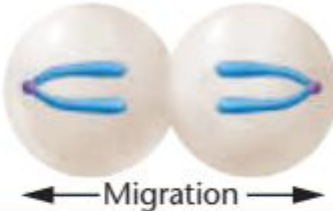
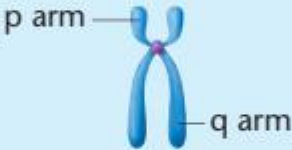
Metaphase

DNA is coiled as
chromosomes

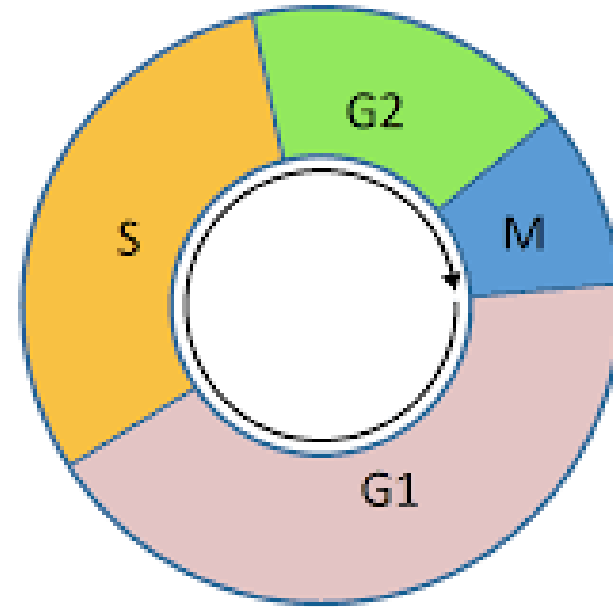
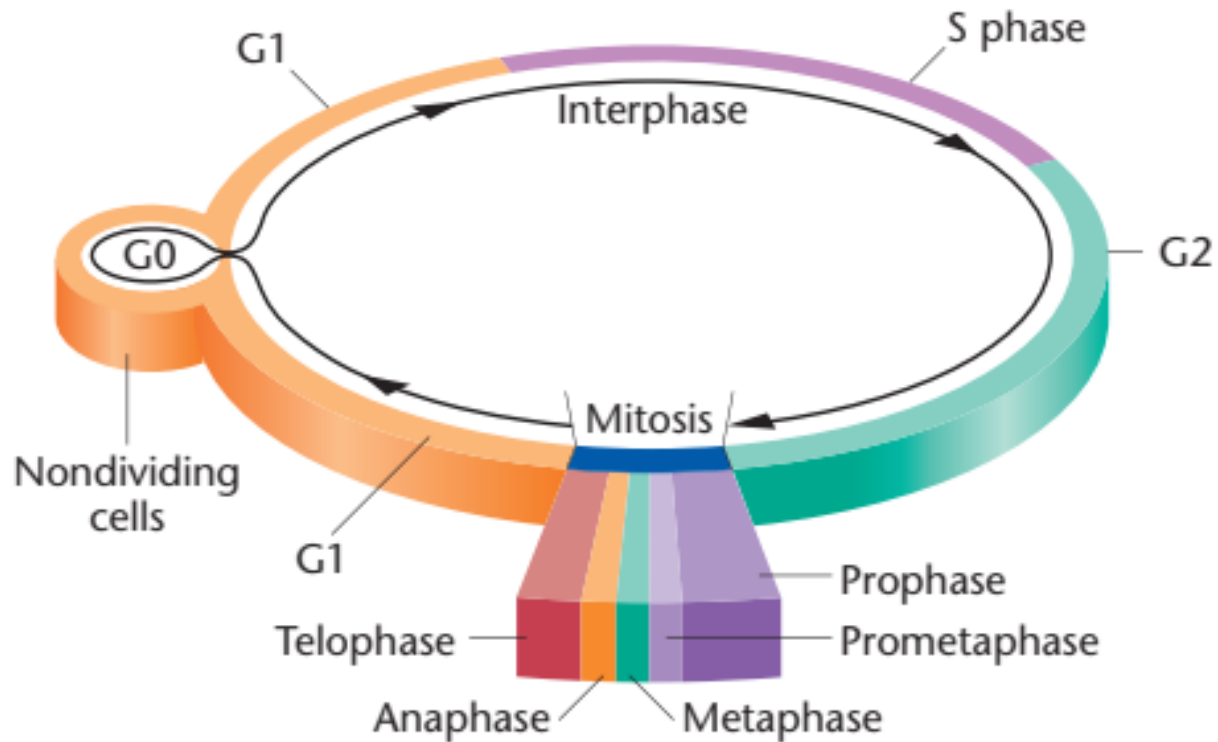




Different shapes of chromosomes

Centromere location	Designation	Metaphase shape	Anaphase shape
Middle	Metacentric	 Sister chromatids Centromere	 Migration
Between middle and end	Submetacentric	 p arm q arm	
Close to end	Acrocentric		
At end	Telocentric		

Stages of the cell cycle



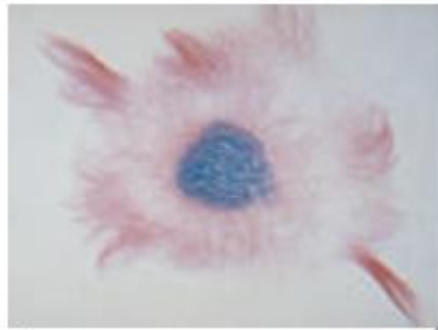
G1 - Growth

S - DNA synthesis

G2 - Growth and preparation for mitosis

M - Mitosis (cell division)

Stages of mitosis



Interphase

Chromosomes are extended and uncoiled, forming chromatin



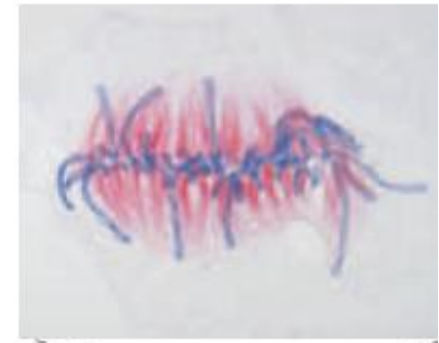
Prophase

Chromosomes coil up and condense; centrioles divide and move apart



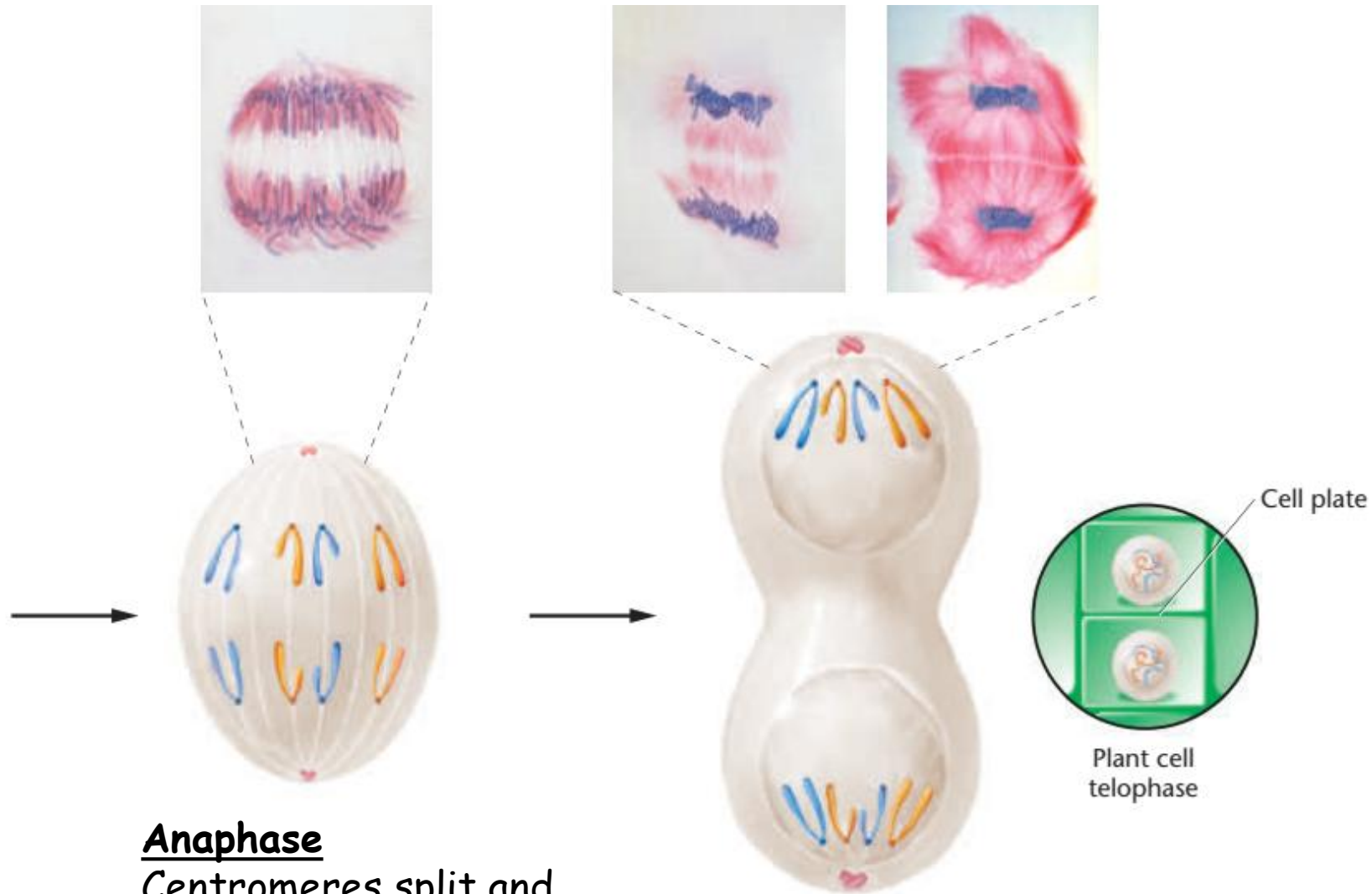
Prometaphase

Chromosomes are clearly double structures; centrioles reach the opposite poles; spindle fibers form



Metaphase

Centromeres align on metaphase plate

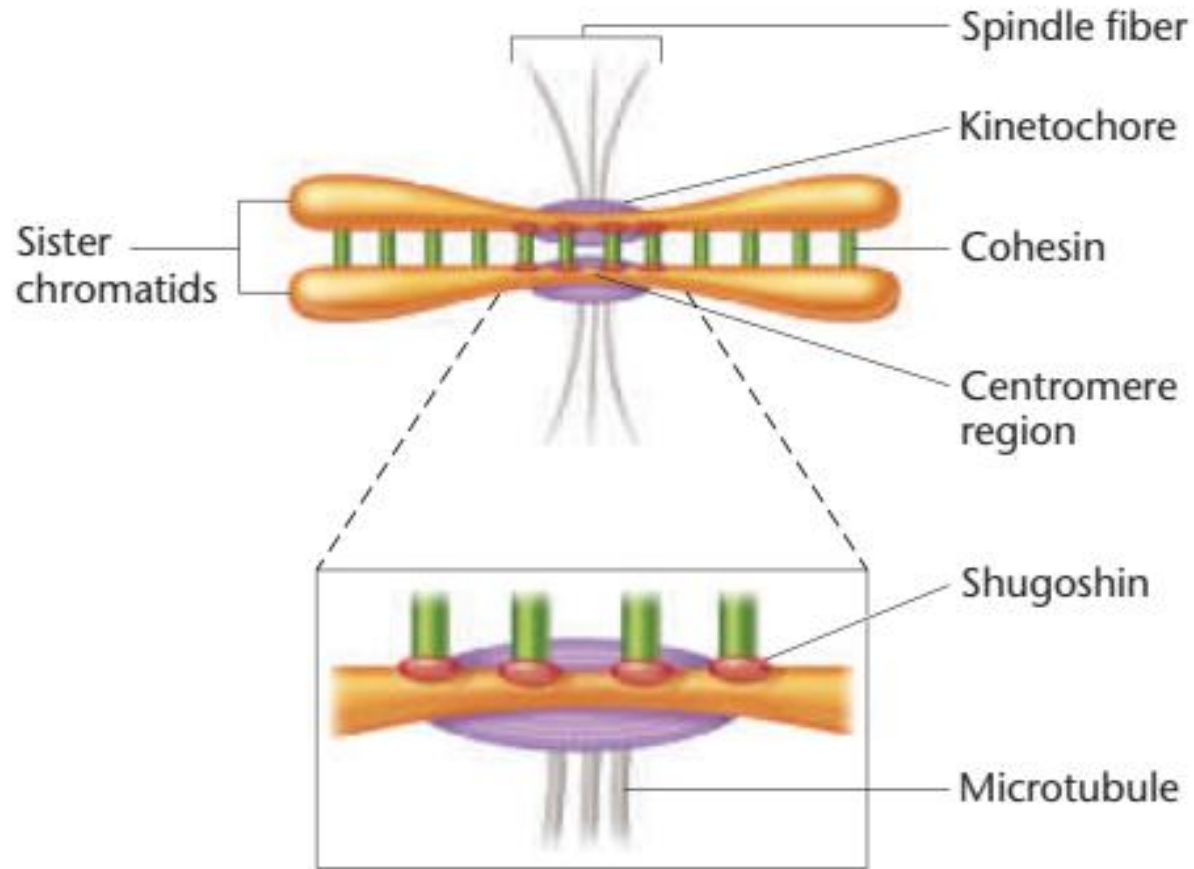


Anaphase

Centromeres split and daughter chromosomes migrate to opposite poles

Telophase

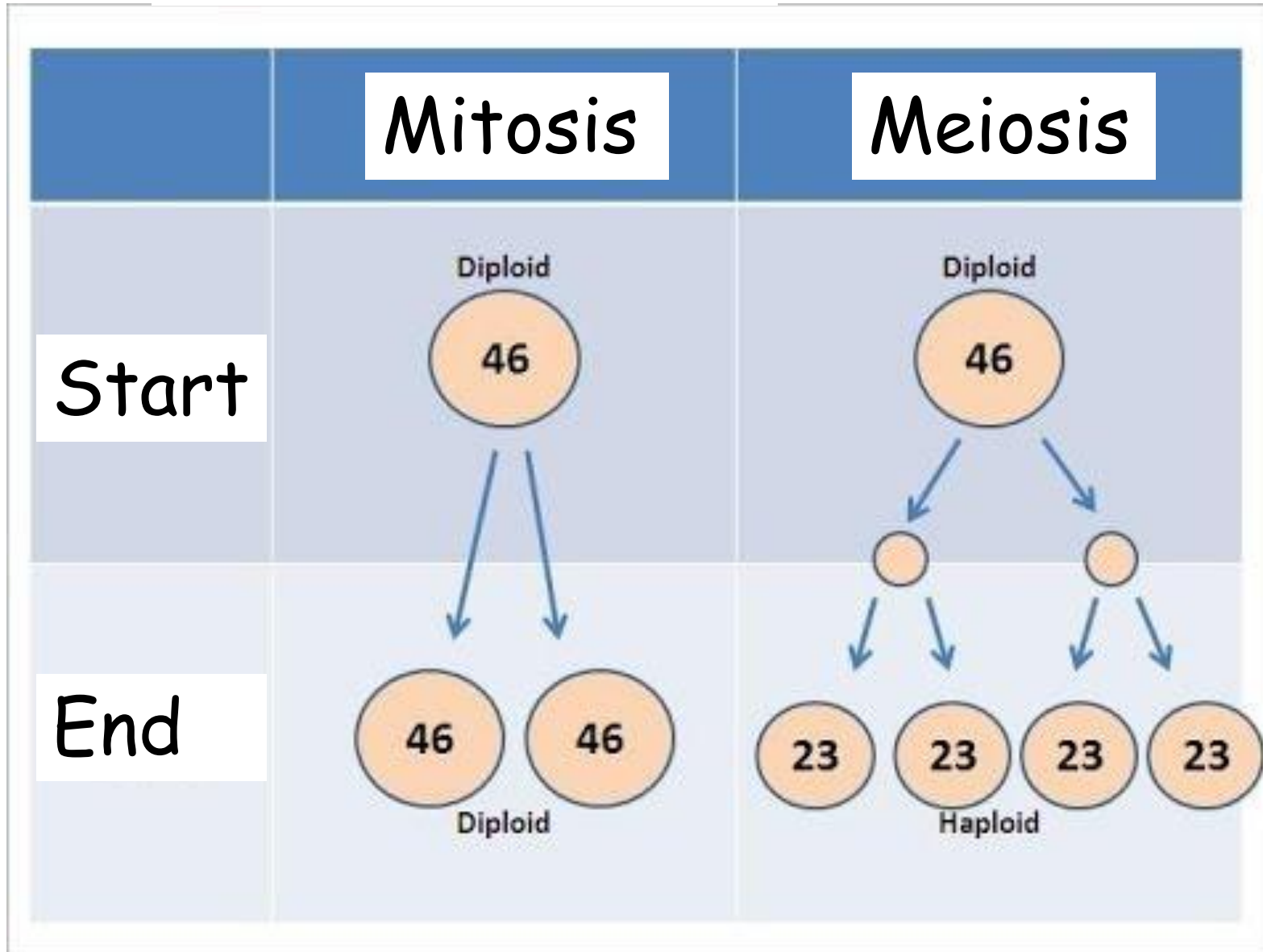
Daughter chromosomes arrive at the poles; cytokinesis commences



Molecular motors
How do
chromosomes
move along the
spindle fibers?

Somatic cells

Germ cells

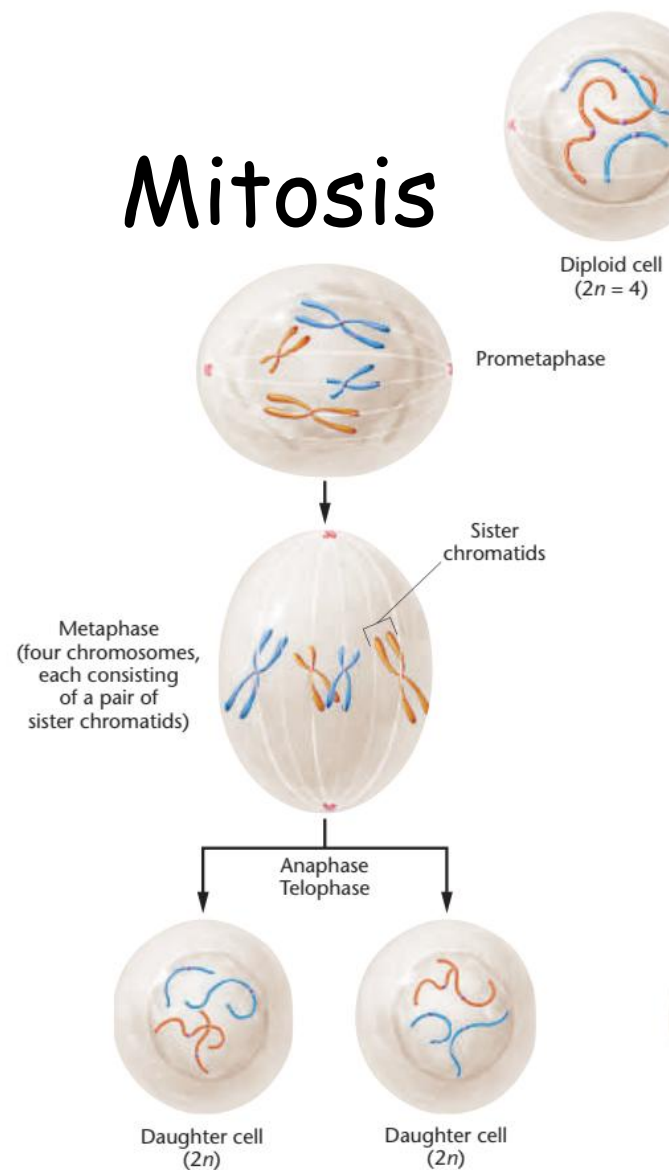


2 divisions

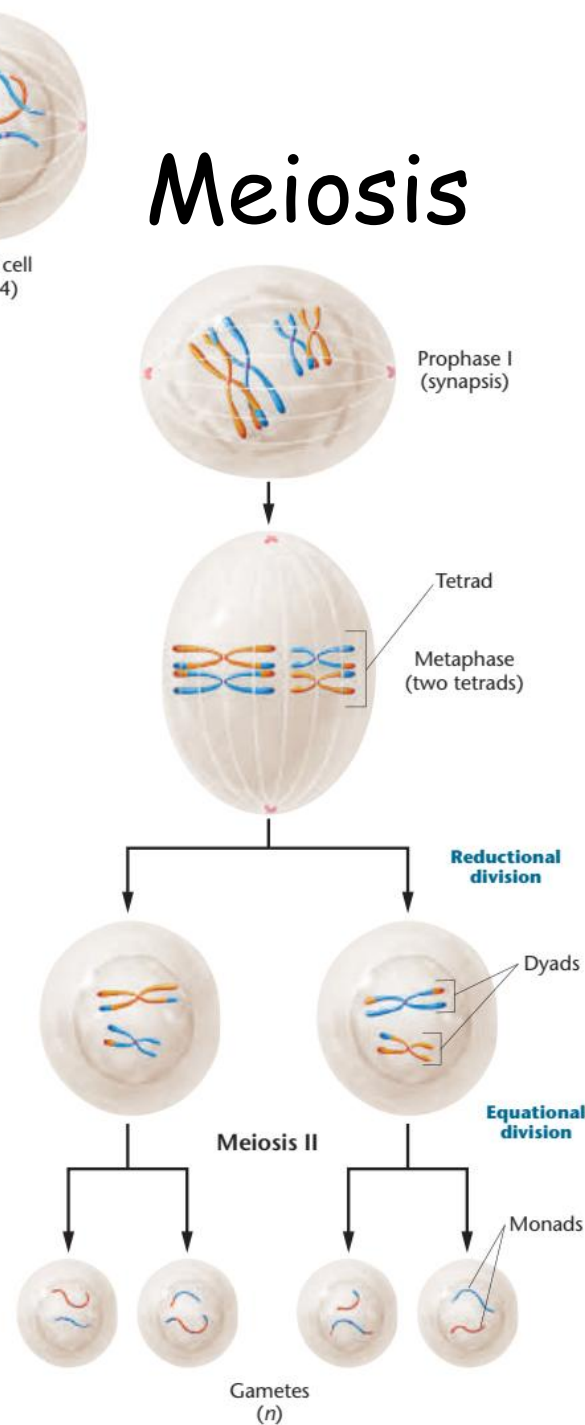
Meiosis I

Meiosis II

Mitosis



Meiosis

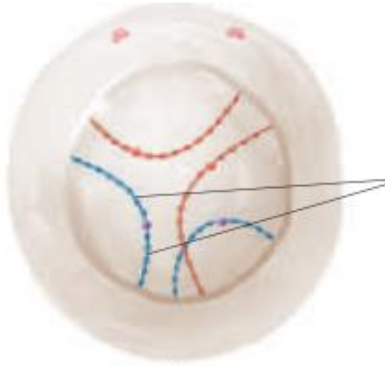


Start
4 chromosomes
($2n$)

End
2 chromosomes
(n)

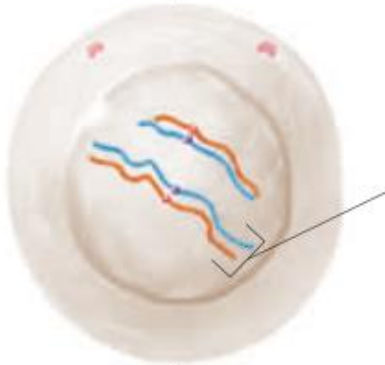
Sub-stages of Meiosis I

Leptonema



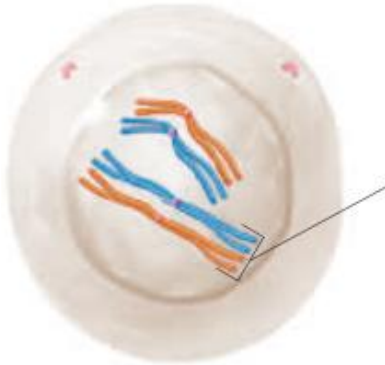
Chromomeres

Zygonema



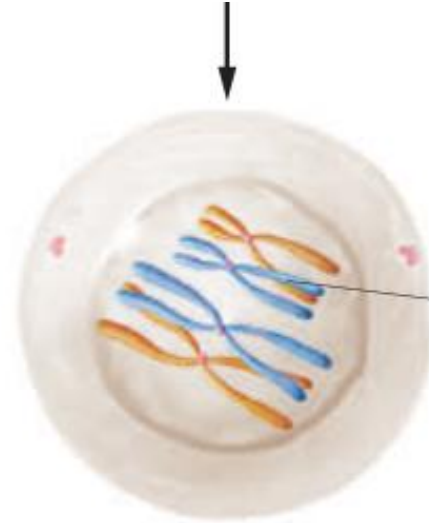
Bivalent

Pachynema



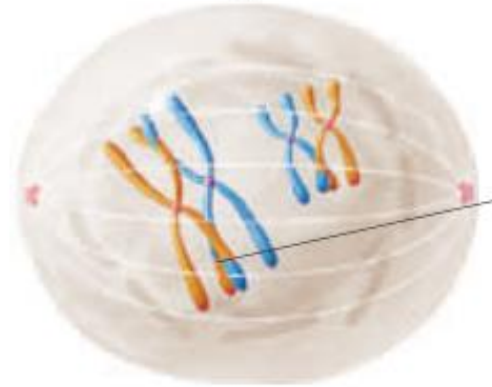
tetrad

Diplonema



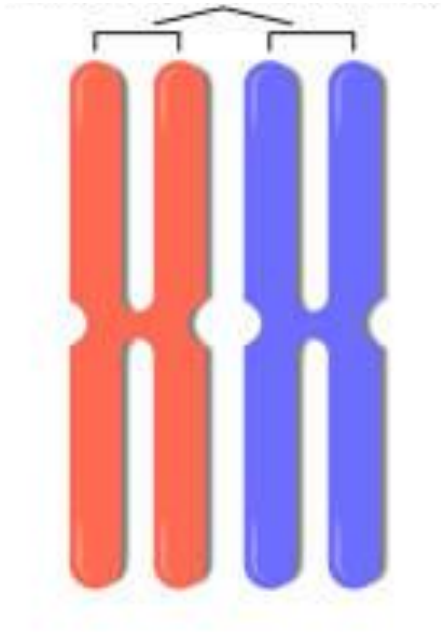
chiasma

Diakinesis

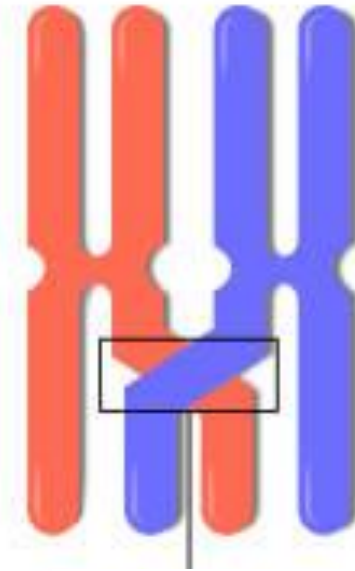


terminal
ization

Homologous
chromosome



Crossing over



chiasma

Recombinant
chromatids



When we were existing as
our father's sperm and
mother's egg, our
grandparents' chromosomes
recombined in this way

Metaphase I



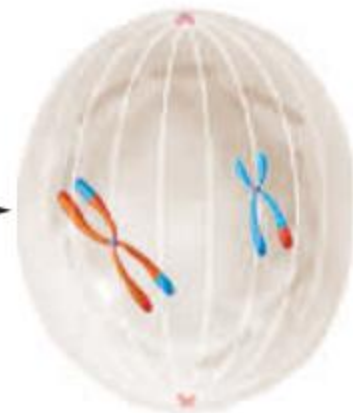
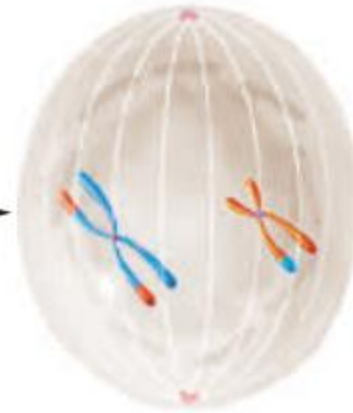
Anaphase I

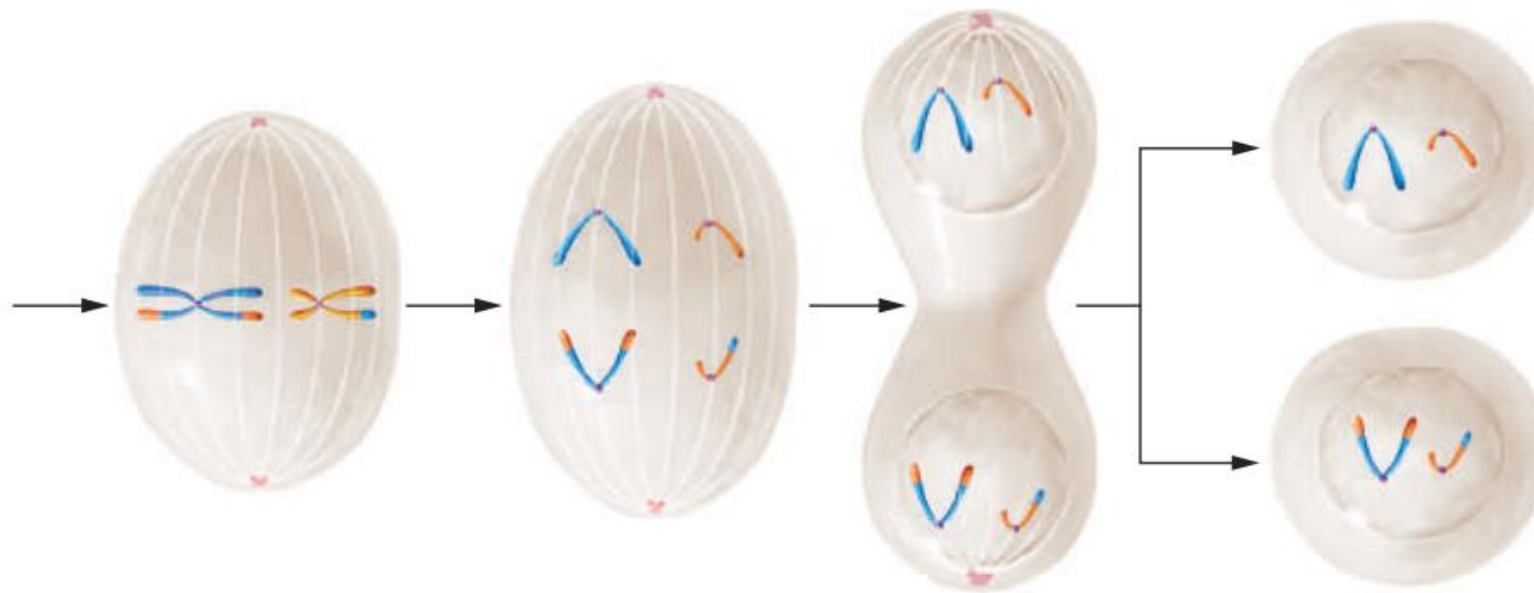


Telophase I

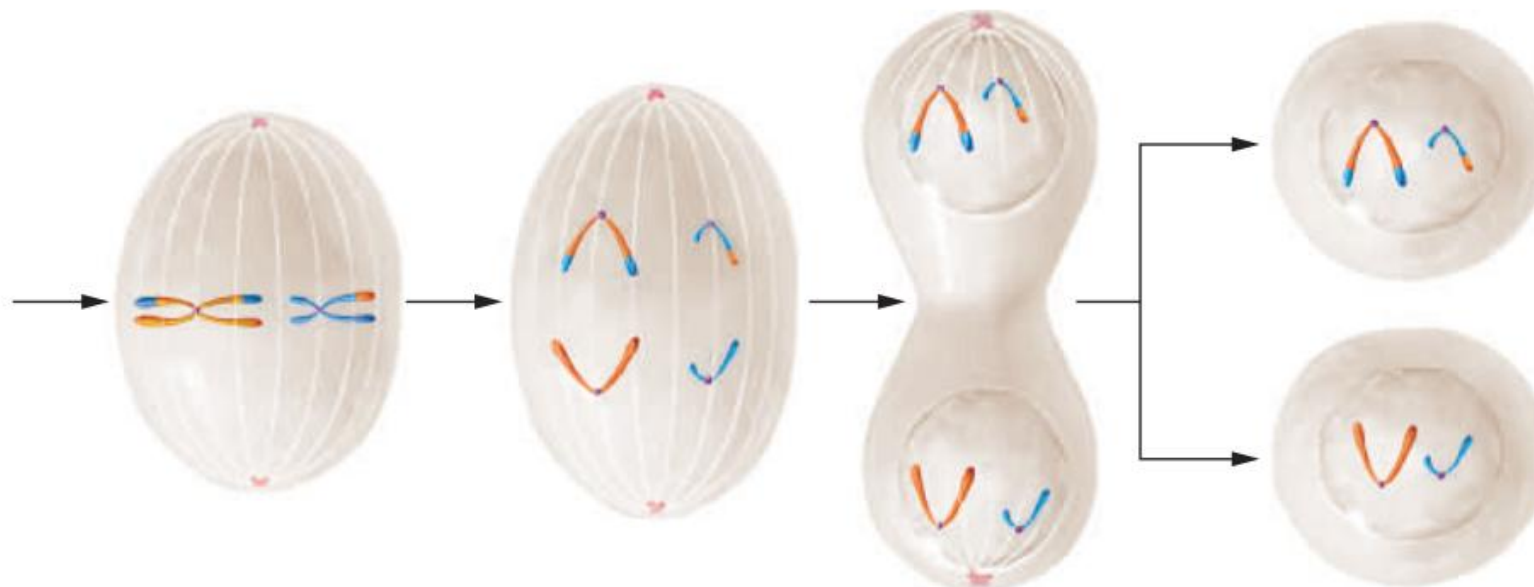


Prophase II





Metaphase II Anaphase II Telophase II Haploid gametes



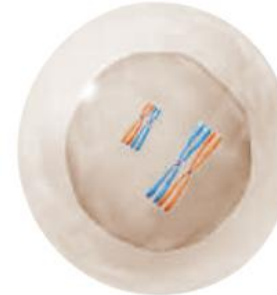
Spermatogonium



Oogonium



Growth/Maturation



Primary spermatocyte

Primary oocyte

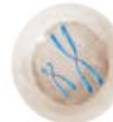
Meiosis I

Secondary spermatocytes

Secondary spermatocytes



Secondary oocyte



First polar body

Meiosis II

spermatid



ootid



Second polar body

Differentiation

spermatozoa

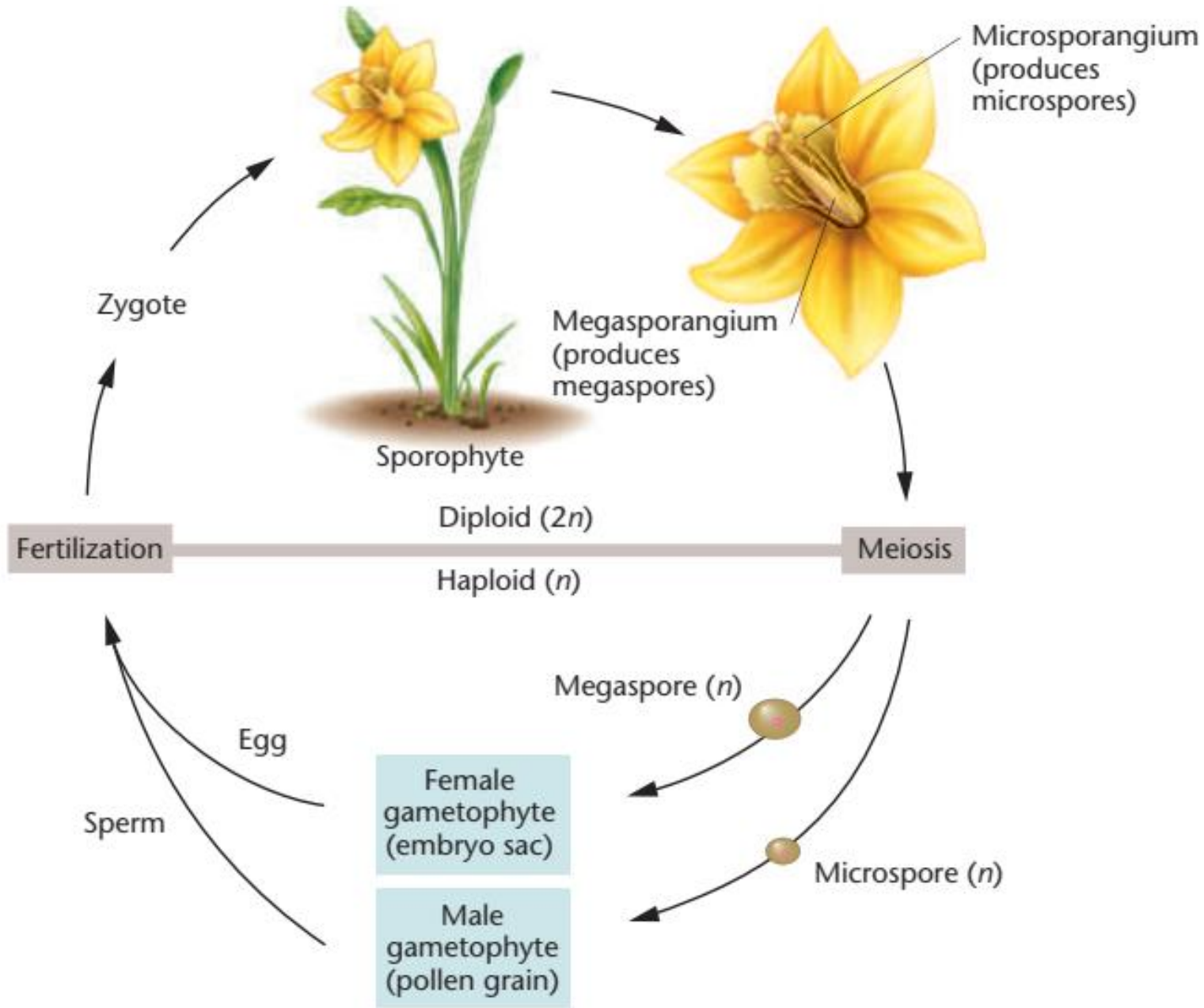


Ovum



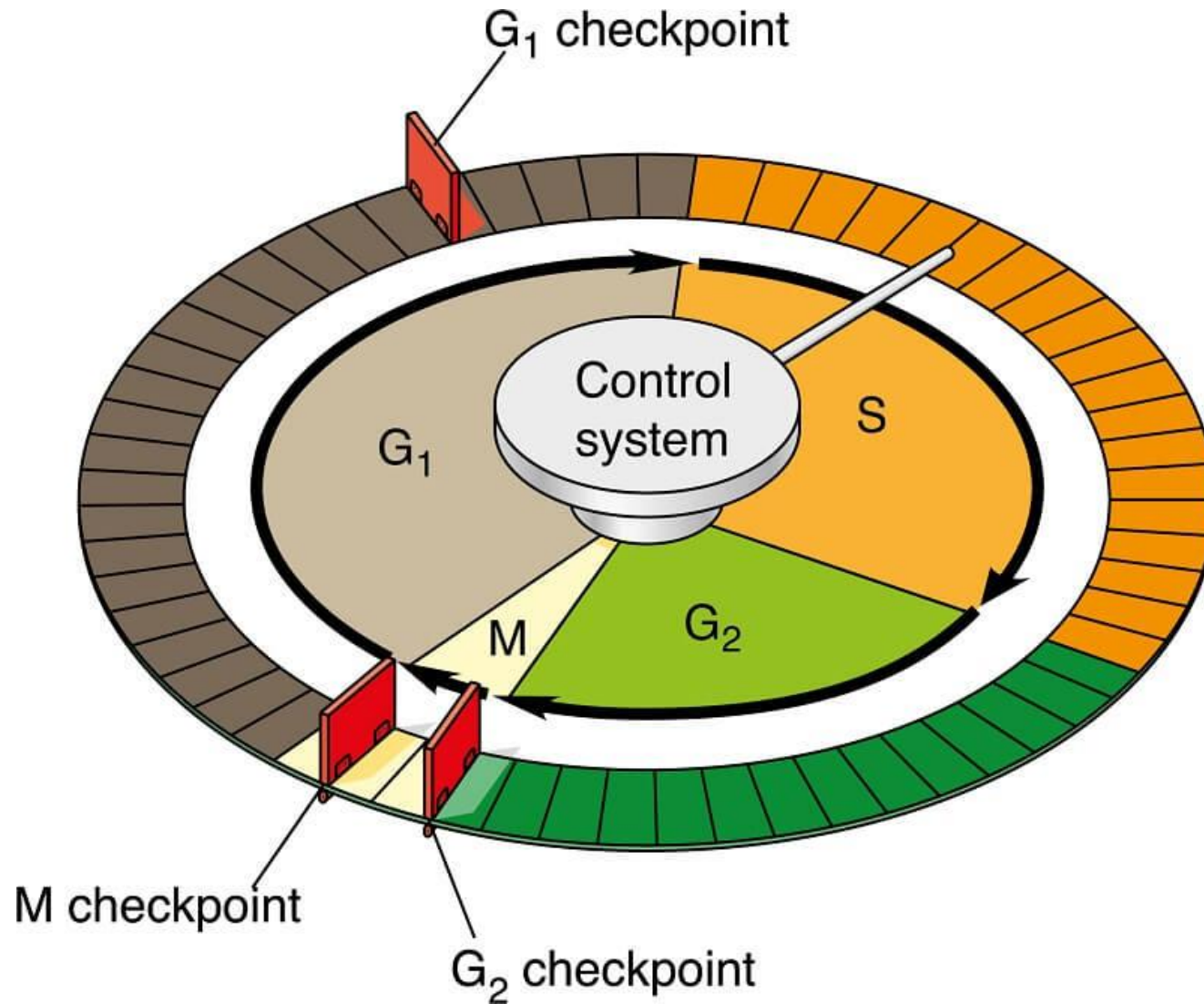
When girls are within their mother's womb, their own egg production (Meiosis I) starts!

Meiosis II continues in puberty



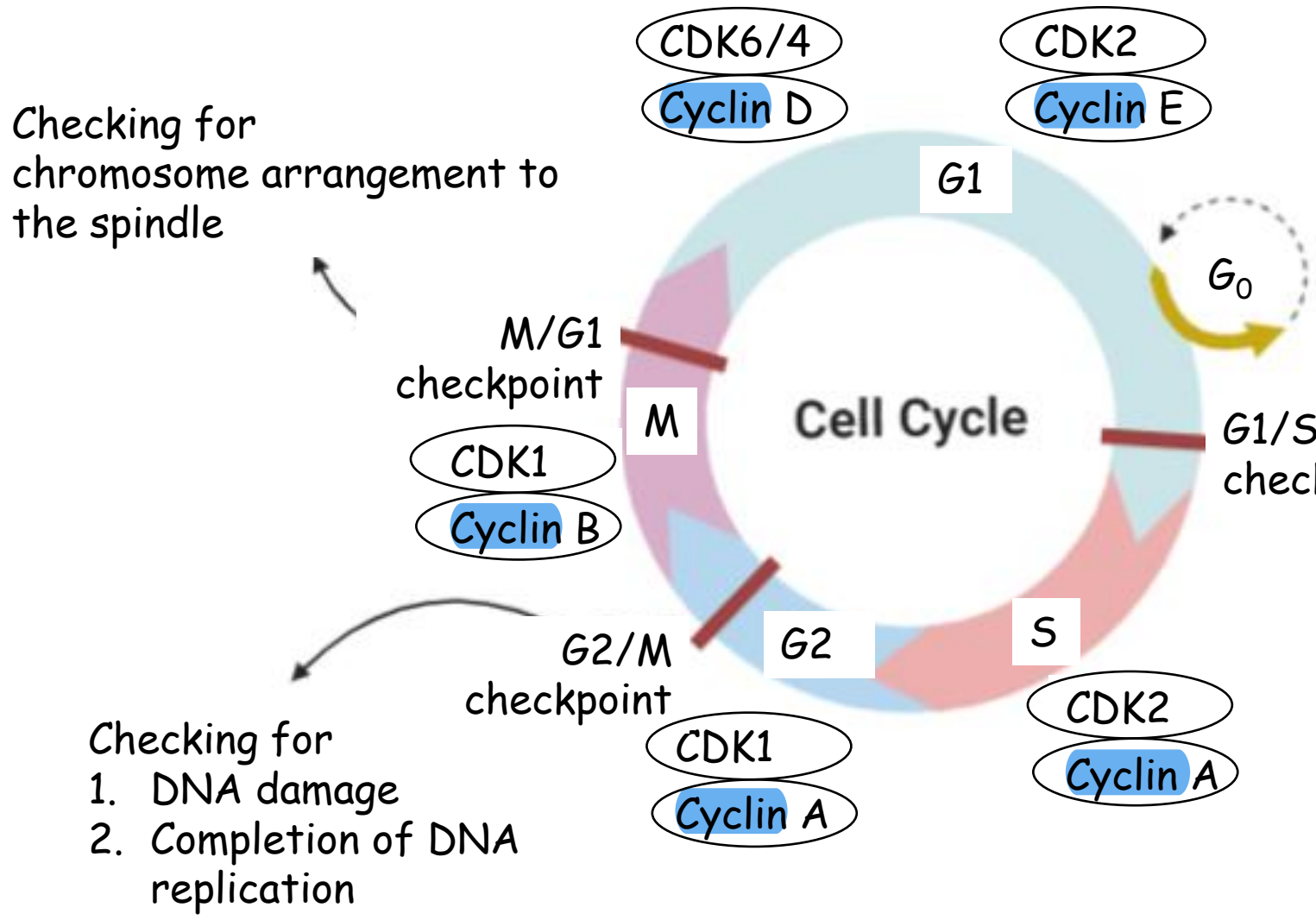
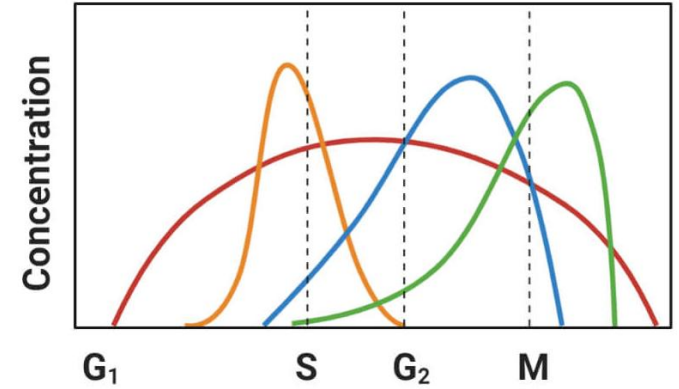
Alternation of generations
in plants

Haploid $\rightarrow$ diploid $\rightarrow$ haploid



Regulation of
the cell cycle

'checkpoints'



Checking for chromosome arrangement to the spindle

Checking for
1. DNA damage
2. Completion of DNA replication

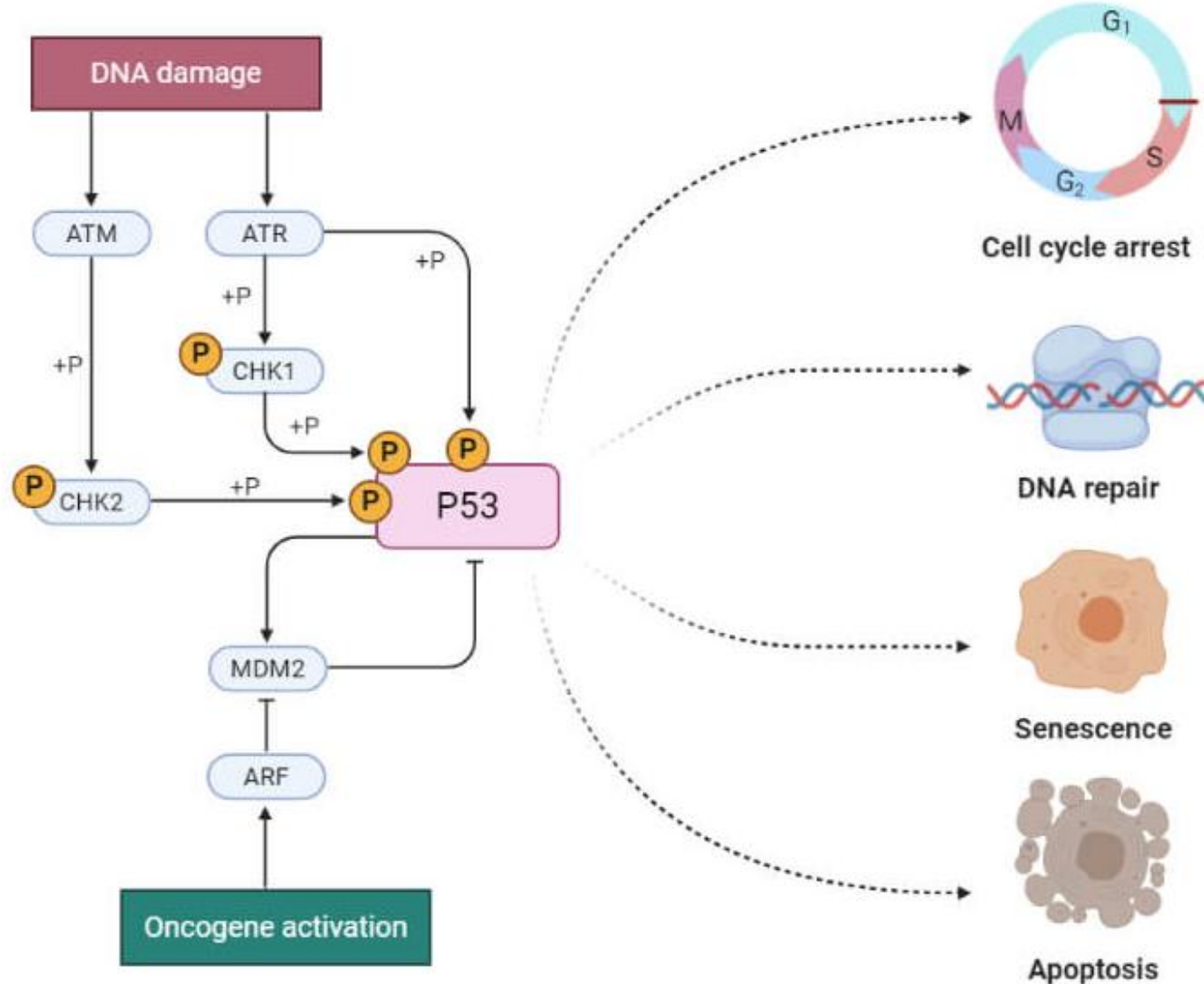
G1/S checkpoint

Checking for
1. Cell size
2. Nutrients and energy
3. DNA damage
4. Growth factors

Cyclins and cyclin-dependent kinase (CDK)

P53 - tumor suppressor gene

P53: the guardian of the genome



- P53 ensures proper cell cycle as it prevents the division of cells with damaged DNA.
- Concentration of p53 in a normal cell is quite low; however, it increases due to DNA damage or stress signals.
- The p53 gene can perform one of three functions, cell cycle arrest, DNA repair, and **apoptosis**.

Failure of regulation of the cell cycle leads to **cancer**